

Conclusion: Elevating Language Models from
Vectors to Enriched Category Frameworks

What This Paper Actually Did: Eleven Concrete Deliverables I

This paper developed a unified category-theoretic framework for linguistic case systems, synthesizing **five formal traditions**—typological, type-logical, distributional, enriched-categorical, and topos-theoretic—together with a **sixth strand**, the biolinguistic and neurocomputational interface (ROSE and related oscillatory accounts; ??, ??), and embedding the whole within an active inference model of cognition. Our eleven principal contributions are:

Notation: Each entry is labelled **C_n** (*nth contribution, C1–C11*) or **F_n** (*nth future direction, F1–F8*).

What This Paper Actually Did: Eleven Concrete Deliverables II

C1: Case Categories as a Formal Algebraic Framework

The review formalized case systems as categories with case roles as objects and grammatical relations as morphisms (??). This formalization captures the full typological range of alignment systems—nominative-accusative, ergative-absolutive, active-stative, tripartite, and fluid-S—within a single algebraic framework, with alignment functors providing structure-preserving mappings between systems. By tracing the lineage from Fillmore's [-@fillmore1968case] deep cases through Jakobson's binary distinctive features and Dowty's [-@dowty1991thematic] proto-roles, the analysis demonstrated how enriched (weighted) morphisms link the categorical formalization to the gradient nature of thematic role assignment.

What This Paper Actually Did: Eleven Concrete Deliverables III

C2: String Diagrams for Case Derivation Visualization

Building on Joyal and Street's [JoyalStreet1991geometry] string diagram formalism and its application in categorial grammar (??), we showed how case-marked noun phrases receive type-logical assignments that are fully visualizable as string diagrams. The Curry–Howard correspondence ensures that syntactic well-formedness guarantees semantic compositionality, and the diagrammatic format provides Shimojima's [Shimojima1996reasoning] “free ride” inferences—conclusions about argument structure that are perceptually available from the diagram without explicit computation. We further demonstrated that passivization reduces to a *type permutation* (a Swap operation in the pregroup category), making voice alternation visible as a topological feature of the string diagram.

What This Paper Actually Did: Eleven Concrete Deliverables IV

C5. Case-Marked DisCoCat, the Distributional–Formal Synthesis, and Discourse Extension

We extended the DisCoCat framework (??) with case-typed noun spaces and alignment-sensitive meaning functors, and showed how the recent DisCoCirc [defelice2022discocirc] and QNLP [lorenz2021lambeq] developments extend this analysis to discourse-level structure and quantum hardware respectively. We formalized the *compact closure axiom* (snake equation) that underpins pregroup reductions, demonstrating that the cup-cap zigzag identity provides a visual coherence proof with genuine cognitive significance. Diagram complexity metrics—normal form and depth—provide a quantitative bridge between the type-logical and distributional perspectives on linguistic structure. A key contribution is the demonstration that DisCoCat constitutes the *algebraic formalization* of the distributional programme that modern large language models—from Word2Vec [mikolov2013efficient] through BERT [devlin2019bert] and GPT [radford2018improving]—implement empirically: the categorical

Eight Open Directions I

The following eight directions (**F1–F8**) identify the most tractable and impactful extensions of this framework:

Eight Open Directions II

F1: Computational Experiments with DisCoCirc and lambeq

The DisCoCirc framework [defelice2022discocirc] offers a natural platform for testing our case-theoretic predictions computationally. The release of **lambeq Gen II** [lambeq2025genii] with full DisCoCirc support makes this direction immediately tractable: discourse-level case role tracking—including the dynamic role reversals of ??—can now be compiled into parameterized quantum circuits (PQCs) and trained end-to-end. Krawchuk et al. [krawchuk2025paramqnlp] demonstrate efficient generation of PQCs from large-scale texts (up to 6,410 words) with competitive performance on sentiment classification; [letcher2024tight] and [rad2024trainability] provide gradient bounds and reduced-domain initialization techniques that mitigate barren plateaus, making discourse-level circuit training practically feasible. Concrete experiments could include:

- ▶ Implementing case-marked DisCoCat models in **lambeq Gen II** and evaluating them on semantic role labeling (SRL) tasks, leveraging the discourse-level wiring to track case roles across sentence boundaries

Five Takeaways: One Argumentative Line per Formal Pillar

I

For the reader who absorbs nothing else, the argumentative line reduces to these five points — one per pillar of the formal architecture:

1. **Case is category-theoretic, not morphological.** The universal fact of linguistic case is that every language wires up *who did what to whom*; the formal content of that wiring is precisely the data of a category with case roles as objects and grammatical relations as morphisms. Alignment typologies (nominative, ergative, tripartite, active-stative, fluid-S) are structure-preserving functors between case categories — a typological taxonomy becomes a proof-by-functor.

Five Takeaways: One Argumentative Line per Formal Pillar II

2. **String diagrams make the grammar executable.** Lambek pregroup types, compact closure, and the snake equation compile each sentence into a DisCoPy string diagram whose normal form is the sentence type s ; each reduction is a proof of well-typedness, and the reduction *is* the diagram. The three-sentence DisCoCirc example ships a working discourse with per-entity role-history ribbons showing Alice traversing $\text{NOM} \rightarrow \text{ACC} \rightarrow \text{NOM}$.
3. **Enriched weights ground graded intuitions.** Moving from **Bool**- to $[0, 1]$ -enrichment replaces binary admissibility with graded distributional proximity, makes categorical magnitude $|\mathcal{C}| \approx 2.5$ a computable summary of effective role distinctions, and gives the enriched weight $w_f = \mathcal{C}(A, B)$ used throughout ?? / ?? as the precision on every prediction error.

Five Takeaways: One Argumentative Line per Formal Pillar III

- 4. DAIF ports the whole scaffolding into active inference.** Variational message passing, the Bethe free energy, and a four-term expected free energy $G(\pi)$ turn distributional case assignment into a first-principles inference loop. The N400 and P600 ERP amplitudes fall out of a first-order expansion $\Delta F \approx -\Delta \mathbb{E}[Z] + \frac{1}{2} \Delta \Lambda \sigma_Z^2$ — derivations, not empirical ansätze.
- 5. The same formal layers give AI-safety researchers three concrete handles.** Type discipline on messages, decidable admissibility of multi-turn sequences, and graded-confidence attenuation (??) give prompt injection a type-theoretic reformulation (??) — an engineering specification for agent protocols, not a guarantee on untyped present-day LLM APIs.

What the Paper Does *Not* Claim: Consolidated Limitations

Four limitations recur across the framework and are recorded here explicitly rather than left implicit in their sections of origin:

- ▶ **The mean-field approximation in `push_forward_return`** maintains one belief-weighted return distribution instead of per-state distributions $Z(s)$. By the mean-field bound recorded in ?? the approximation error is at most $\gamma \cdot R_{\max} \cdot H[q]$ in the W_1 metric — tight for sharp posteriors, linearly degrading as entropy grows.
- ▶ **The enriched-categorical unification is a conjecture.** Distributional semantics, distributional RL, and active-inference posteriors all instantiate $[0, 1]$ -enriched structures, but a strict categorical proof that the three share a common enriching monoidal base remains open (??).

What the Paper Does *Not* Claim: Consolidated Limitations II

- ▶ **Empirical validation is narrow.** Case-assignment demonstrations use a single German sentence. Cross-linguistic and cross-register generalisation is left to future work; the hooks in `make_daif_belief_trajectory_data()` make adding corpora a one-function change.
- ▶ **ERP amplitudes are not calibrated to V .** The DAIF predictions are on a dimensionless return/log-probability scale; converting to V would require a per-subject scaling constant fit to empirical ERP data. The qualitative ordering and graded precision response are predictions the current framework *does* make.

Anticipated Objections and Responses I

Four lines of objection stand out; we address each honestly rather than waiting for a reviewer.

Objection 1 — “The mean-field approximation throws away the whole point of distributional RL.” *Response.* The belief-weighted collapse gives $\mathcal{O}(n)$ memory in place of $\mathcal{O}(n \cdot N_{\text{atoms}})$ and is exact in the sharp-posterior limit. The mean-field bound recorded in ?? quantifies the degradation as $H[q]$ grows. The alternative — maintaining per-state return distributions throughout a linguistic parse — would multiply the runtime of `distributional_case_assignment()` by a factor of n (the role count) for a gain that vanishes as the posterior sharpens after the first few words of a sentence.

Anticipated Objections and Responses II

Objection 2 — “The enriched-categorical unification is only stated, never proven.” *Response.* Conceded, and flagged in ?? and in the What’s New block of ?. Distributional semantics, distributional RL, and active-inference posteriors all *instantiate* $[0, 1]$ -enriched structure; whether they share a common enriching monoidal base in the strict sense of Kelly enriched-category theory is an open question. The repository’s tests verify the enriched axioms hold in each instance; they do not construct the common base functor, and the manuscript nowhere uses the strict unification as a load-bearing step in any downstream argument.

Anticipated Objections and Responses III

Objection 3 — “Cross-linguistic evidence is a single German sentence.” *Response.* The empirical scope reflects the publication’s focus on *specification* rather than *corpus study*. The `Discourse.role_reversal("Alice", "Bob")` and `make_daif_belief_trajectory_data()` interfaces accept arbitrary lexical heads and observation sequences; extending to Basque, Dyirbal, Russian, Serbian/BCS, or any other language is a one-function change. ?? (in ??) already shows a pregroup multilingual isomorphism across English / Latin / Japanese, and the Slavic discussion in ?? extends the same type calculus to overtly case-marked Russian and Serbian noun phrases. Slavic-language ERP datasets — Bornkessel-Schlesewsky and colleagues’ eADM-aligned Russian case-violation paradigms in particular [bornkessel2006extended] — would provide a near-term empirical test for the precision-weighted P600 prediction in **C10 / F4**. The three-sentence Alice/Bob discourse is a proof-of-concept, not a typological claim.

Anticipated Objections and Responses IV

Objection 4 — “The ROSE phase–amplitude coupling gap means you cannot predict ERP latencies.” *Response.*

Conceded and flagged explicitly in ???. The current implementation keeps N400 and P600 *peak latencies* as fixed Gaussian peaks at 380 ms and 600 ms respectively (see `DEFAULT_N400_PEAK_MS / DEFAULT_P600_PEAK_MS` in `src/daif/prediction.py`). A principled latency prediction would require a cross-frequency-coupling delay parameter inside the CEREBRUM layer; we flag this as the cleanest next research step.

What to Read Next, by Reader Profile I

Readers who entered this paper from different fields will want different onward paths:

- ▶ **Linguists / typologists:** skip ahead to ??–?? for the categorical formalisation of case and the five alignment functors; then ?? for the complexity metrics that make cross-linguistic comparison quantitative.
- ▶ **Machine-learning researchers:** ?? first for the Distributional Active Inference extension of the Bellemare-Dabney-Munos programme to linguistic case; then ?? for how DAIF yields falsifiable ERP predictions.
- ▶ **Cognitive / computational neuroscientists:** ?? for the active-inference framing; ?? for the three falsifiable ERP predictions; ?? for the PAC-latency gap that is the cleanest experimental entry point.

What to Read Next, by Reader Profile II

- ▶ **QNLP / quantum-semantics engineers:** ?? and ?? for the POVM-based case assignment and the honest separation of implemented POVM machinery from literature-bridge TQNN / ZX / lambeq claims.
- ▶ **AI-safety / agent-protocol practitioners:** ?? for the three concrete handles (type discipline, decidable admissibility, graded confidence) and ?? for prompt injection as a type violation — engineering specification rather than automatic guarantee on today's APIs.

Case Categories Are the Geometry of Meaning: A Unifying Coda I

The commutative diagram is the central motif of this review—both as a mathematical tool and as a cognitive instrument. The analysis has demonstrated that the same diagrammatic language that makes category theory effective for formalizing case systems also makes it effective for *thinking about* case systems: the spatial structure of a commutative diagram encodes relational information in a format that supports rapid search, pattern recognition, and free-ride inference.

Case Categories Are the Geometry of Meaning: A Unifying Coda II

This convergence of mathematical utility and cognitive efficacy is not accidental. If the active inference framework is correct, then the brain operates by constructing and updating generative models of the world's relational structure. Category theory provides the structural algebra for these generative models; commutative diagrams supply their natural topology; and case categories instantiate the precise relational vocabulary that cognitive systems deploy to organize experience into coherent narratives of *who does what to whom*. The distributional revolution in both semantics and reinforcement learning—from Firth's [-@firth1957papers] co-occurrence statistics through transformer attention weights to Distributional Active Inference [@akgul2026distributional]—confirms that meaning is not an atomic property of symbols but emerges from the relational structure of contexts, a principle that enriched category theory captures with mathematical precision.

Case Categories Are the Geometry of Meaning: A Unifying Coda III

Crucially, **this synthesis clarifies a mathematical angle on alignment for relational agent cognition.** Moving from flat token streams toward **explicitly typed** enriched categorical interaction grammars lets one state **when** prompt injection corresponds to a **type error relative to a fixed protocol**—the conditional analysis in **??**. Default LLM stacks do not yet enforce such protocols end-to-end; the payoff is sharper **specification** (what boundary checks would guarantee) and a research agenda for non-cartesian wiring, not a blanket claim that injection is already impossible in production systems.

Case Categories Are the Geometry of Meaning: A Unifying Coda IV

Ultimately, the mathematics of case alignment presents a highly structured formal geometry of meaning—the relational algebra through which cognitive agents navigate and render intelligible the structured world of events, participants, and relations. The convergence of formal semantics, distributional semantics, and active inference within a single categorical framework suggests that commutative diagrams offer a natural formalism in which relational cognition can be modeled.